



INTERNATIONAL MATHEMATICS SUMMER CAMP IMSC25
MOCK TEST 2-COMBINATORICS

Date: Thursday, 26th June 2025
Number of problems: 3

Time: 13:30-18:00
Total points: 21

Problem 1. A king rules a country with 2026 citizens (including himself). Each citizen has a salary of 1\$. The king has the authority to propose a law that changes the distribution of salaries, provided that each salary is a nonnegative integer and the total sum of salaries remains 2026\$. The change is approved if more than half of the voters vote in favor, where the king does not vote. People are selfish: everyone votes in favor if his salary increases, votes against if his salary decreases, and does not bother to vote if his salary remains the same.

- a) What is the maximum salary that the king can ascertain for himself (possibly after more such law changes)?
- b) How many (at least) law changes are necessary for the king to reach the maximum salary?

Solution 1. The answer is 2023\$ for the part a), and 11 for the part b).

In the first step, the king proposes for 1013 citizens to have a salary of 2\$, while everybody else has 0\$ (including the king). After that, he chooses 508 lucky citizens who will have their salaries further increased to 3\$ or 4\$, while everybody else will get 0\$ salary. In further turns, the number of citizens who have any salary is reduced to 255, 128, 65, 33, 17, 9, 5, 3, 2, respectively. In the last step, the king gives a salary of 1\$ to some three citizens who had no salary before that, and collects everything else for himself.

Let us prove that he cannot do better. If he could, then for the last voting there would be at most two citizens whose salaries increased from 0\$ to 1\$, that is, at most two voters for the king's proposal. But then there could be at most one vote against, that is, at most one person whose salary has been reduced from nonzero to zero. In other words: there had to be a moment when there was only one citizen (not counting the king) who has a nonzero salary. But it is clear that, in one step, the number of voters who have a nonzero salary can be reduced only to just over half of the number of such voters from the previous step. This means that it is impossible to reach a situation with only one voter with a nonzero salary, and at the same time, this also means that the king cannot achieve his intention in less than 11 steps.

Marking scheme:

- a. 1 point – claiming 2023\$ is the solution.
- b.1 1 point – showing 2023\$ can be obtained
- c. 2 points – proving no more than 2023\$ can be obtained
- b.2 1 point – showing 2023\$ can be obtained in 11 steps
- d. 2 points – proving at least 11 steps are necessary

Problem 2. A finite set of integers, $S \subset \mathbb{Z}$, is called *good* if every $s \in S$ can be written as a sum of some two elements (not necessarily distinct) from S . Such a set has *safety level* n if it does not contain a subset of n or less elements whose sum is 0. Does there exist a good set with safety level 2025?

Solution 2. Let $a_i = 2^i$ and $b_i = -(2^{n+1} - 1 - 2^i)$, where $n \geq 2025$. We claim that the set $\{a_i, b_i : 0 \leq i \leq n\}$ has safety level n (and thus safety level 2025). First, it is clear that it satisfies the requirement about representability of each element by sum of other elements: $a_i = a_{i-1} + a_{i-1}$ for each $i \geq 1$ while $a_0 = a_n + b_n$, and $b_i = b_{i-1} + a_{i-1}$ for each $i \geq 1$ while $b_0 = b_n + b_n$. We are now going to find the subset of minimal cardinality in which the total sum of elements is 0. Having in mind that all the numbers a_i are positive and all the numbers b_i are negative, we note that any such subset can contain at most one of the numbers b_i : indeed, $b_n + b_{n-1}$ (which are two elements with the smallest absolute value) $= -(5 \cdot 2^{n-1} - 2)$, and this negative amount cannot be compensated even by all the numbers a_i together (their sum is $2^{n+1} - 1 = 4 \cdot 2^{n-1} + 1 - 2 < 5 \cdot 2^{n-1} - 2$). Therefore, the considered subset consists of a particular b_k and some of the numbers a_i which sum to its absolute value. However, we clearly have

$$2^{n+1} - 1 - 2^k = \sum_{\substack{i=0 \\ i \neq k}}^n 2^i = \sum_{\substack{i=0 \\ i \neq k}}^n a_i,$$

and thus in the considered subset, beside b_k , there are n more of the numbers a_i , which altogether makes $n + 1$ elements. This completes the solution.

Marking scheme:

- a. 2 points – giving a correct set
- b. 2 points – proving the set is good
- c. 3 points – proving the set has safety level 2025

Problem 3. A descent of a permutation $\pi = (\pi_1, \pi_2, \dots, \pi_n)$ of $\{1, 2, \dots, n\}$ is a value $1 \leq i \leq n-1$ for which $\pi_i > \pi_{i+1}$. Let $\text{des}(\pi)$ be the number of descents of π . The square of a permutation π is defined by $\pi^2 = (\pi_{\pi_1}, \pi_{\pi_2}, \dots, \pi_{\pi_n})$. Calculate the value

$$\sum_{\pi} \text{des}(\pi^2),$$

where the sum ranges over all permutations π of the set $\{1, 2, \dots, 2024\}$.

Solution 3. We start proving the following lemma.

lemma For a fixed position $1 \leq i \leq n-1$, among all permutations $\pi \in \mathcal{S}_n$ for which $\{\pi(i), \pi(i+1)\} \neq \{i, i+1\}$, half of them satisfy $\pi^2(i) > \pi^2(i+1)$ and half of them $\pi^2(i) < \pi^2(i+1)$.

proof We can represent a permutation π by drawing a directed graph (with possibly loops) with out- an indegree equal to one (for every vertex), by connecting $j \rightarrow \pi(j)$ for every $1 \leq j \leq n$. For every $b, c \in [n] \setminus \{i, i+1\}$, there are equally many permutations for which $(\pi^2(i), \pi^2(i+1)) = (b, c)$ as there are for which $(\pi^2(i), \pi^2(i+1)) = (c, b)$, since one can just switch the vertex i and vertex $i+1$ in the digraph. Similarly, for every $a \in [n] \setminus \{i, i+1\}$, there are equally many $\pi \in \mathcal{S}_n$ for which $\pi^2(i), \pi^2(i+1)) = (i, a)$ as there are for which $\pi^2(i), \pi^2(i+1)) = (a, i+1)$ (again by switching i and $i+1$), and analogously for $(\pi^2(i), \pi^2(i+1)) = (a, i)$ and $(\pi^2(i), \pi^2(i+1)) = (i+1, a)$. Since $a > i$ if and only if $i+1 > a$, the bijections implied that there are equally many permutations for which π^2 is descending as ascending at position i . $\square$

By the lemma, all we need to do is to count the number of permutations for which $(\pi(i), \pi(i+1)) = (i, i+1)$ and $(\pi(i), \pi(i+1)) = (i+1, i)$. The latter happens if the digraph contains a directed C_4 with i and $i+1$ at distance two from each other. There are $(n-2)!$ ways to do so.

In the other case, we have that i either contains a loop, or swaps with an element b . The same is true for $i+1$ and a different element c . There are now $(n-2)!$ ways where

- both have a loop,
- i has a loop and $i+1$ swaps with some j (different from i and $i+1$),
- $i+1$ has a loop and i swaps with some j ,
- i and $i+1$ swap with some j, k resp.,
- i and $i+1$ swap with each other.

This implies that for $n \geq 3$, there are $4(n-2)!$ many more permutations for which i is not a descent than there are for which it is a descent. This gives $\frac{n! - 4(n-2)!}{2}$ many permutations for which i is a descent. Summing over $n-1$ choices for i , we obtain

$$\sum_{\pi} \text{des}(\pi^2) = \frac{(n-1)n! - 4(n-1)!}{2}$$

Remark 1. Note the latter can also be written as e.g. $n! \left(\frac{n-1}{2} - \frac{2}{n} \right)$.

Marking scheme:

- a. 4 points – lemma
- b. 3 points – finishing the proof